

Free Body Diagrams and Control System Block Diagrams for Mechanical Systems

Control Systems Modeling

June 15, 2025

Contents

1 Introduction

This document presents comprehensive free body diagrams and control system block diagrams for various mechanical systems commonly encountered in control systems engineering. Each system is analyzed with its governing equations, free body diagram, and corresponding control system representation.

2 Servo Motor System

2.1 System Description

A servo motor system consists of a DC motor with moment of inertia J , damping coefficient D , and torque constant K_t . The motor current I generates torque to rotate the shaft.

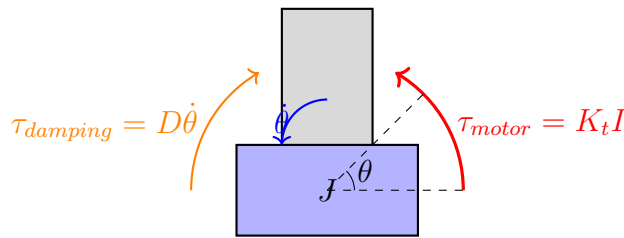
2.2 Governing Equation

$$J\ddot{\theta} + D\dot{\theta} = K_t I \quad (1)$$

where:

- J = moment of inertia ($\text{kg}\cdot\text{m}^2$)
- D = damping coefficient ($\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$)
- K_t = torque constant ($\text{N}\cdot\text{m}/\text{A}$)
- I = motor current (A)
- θ = angular position (rad)

2.3 Free Body Diagram



Servo Motor Rotor

Figure 1: Free Body Diagram of Servo Motor

2.4 Transfer Functions

Position output:

$$G_{\theta}(s) = \frac{\Theta(s)}{I(s)} = \frac{K_t}{Js^2 + Ds} \quad (2)$$

Velocity output:

$$G_w(s) = \frac{s\Theta(s)}{I(s)} = \frac{K_t}{Js + D} \quad (3)$$

2.5 Control System Block Diagram

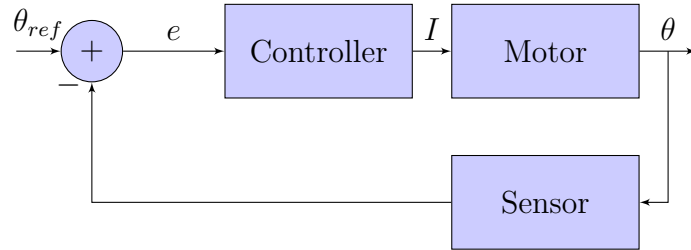


Figure 2: Servo Motor Control System Block Diagram

3 Robotic Arm Joint System

3.1 System Description

A robotic arm joint with moment of inertia M , friction coefficient B , and spring stiffness k representing joint elasticity.

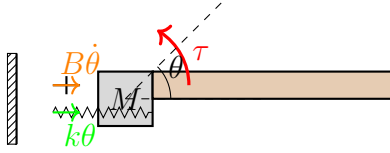
3.2 Governing Equation

$$M\ddot{\theta} + B\dot{\theta} + k\theta = \tau \quad (4)$$

where:

- M = moment of inertia ($\text{kg}\cdot\text{m}^2$)
- B = friction coefficient ($\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$)
- k = spring stiffness ($\text{N}\cdot\text{m}/\text{rad}$)
- τ = applied torque ($\text{N}\cdot\text{m}$)

3.3 Free Body Diagram



Robotic Arm Joint

Figure 3: Free Body Diagram of Robotic Arm Joint

3.4 Transfer Function

$$G(s) = \frac{\Theta(s)}{\tau(s)} = \frac{1}{Ms^2 + Bs + k} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (5)$$

where $\omega_n = \sqrt{k/M}$ and $\zeta = \frac{B}{2\sqrt{kM}}$

3.5 Control System Block Diagram

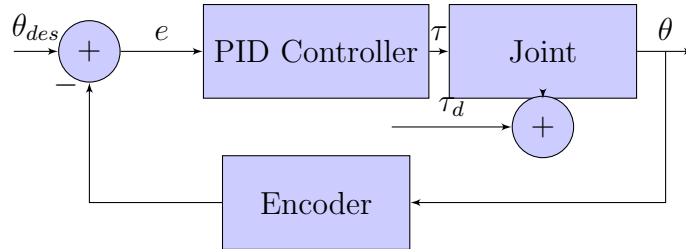


Figure 4: Robotic Arm Joint Control System Block Diagram

4 Ball Screw System

4.1 System Description

A ball screw converts rotary motion to linear motion. The system has mass M , damping B , and experiences axial force F .

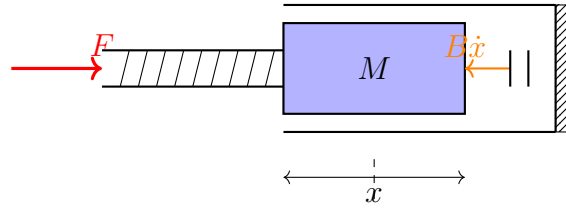
4.2 Governing Equation

$$M\ddot{x} + B\dot{x} = F \quad (6)$$

where:

- M = mass (kg)
- B = damping coefficient (N·s/m)
- F = axial force (N)
- x = position (m)

4.3 Free Body Diagram



Ball Screw Linear Actuator

Figure 5: Free Body Diagram of Ball Screw System

4.4 Transfer Functions

Position output:

$$G_x(s) = \frac{X(s)}{F(s)} = \frac{1}{s(Ms + B)} \quad (7)$$

Velocity output:

$$G_v(s) = \frac{sX(s)}{F(s)} = \frac{1}{Ms + B} \quad (8)$$

4.5 Control System Block Diagram

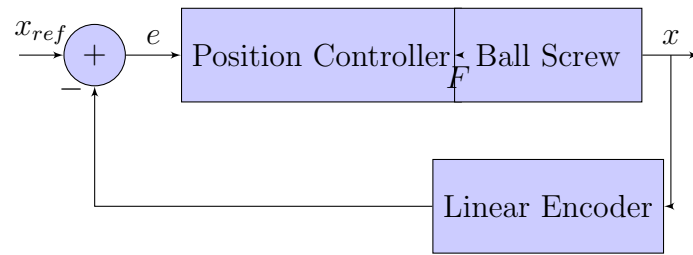


Figure 6: Ball Screw Control System Block Diagram

5 Conveyor Belt System

5.1 System Description

A conveyor belt system with effective mass M , friction B , and drive force F moving materials at velocity v .

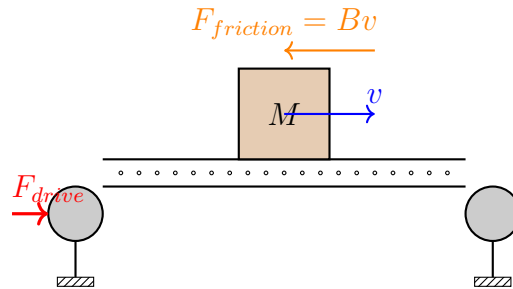
5.2 Governing Equation

$$M\dot{v} + Bv = F \quad (9)$$

where:

- M = effective mass of belt and load (kg)
- B = friction coefficient (N·s/m)
- F = drive force (N)
- v = belt velocity (m/s)

5.3 Free Body Diagram



Conveyor Belt System

Figure 7: Free Body Diagram of Conveyor Belt System

5.4 Transfer Functions

Velocity output:

$$G_v(s) = \frac{V(s)}{F(s)} = \frac{1}{Ms + B} \quad (10)$$

Position output:

$$G_x(s) = \frac{X(s)}{F(s)} = \frac{1}{s(Ms + B)} \quad (11)$$

5.5 Control System Block Diagram

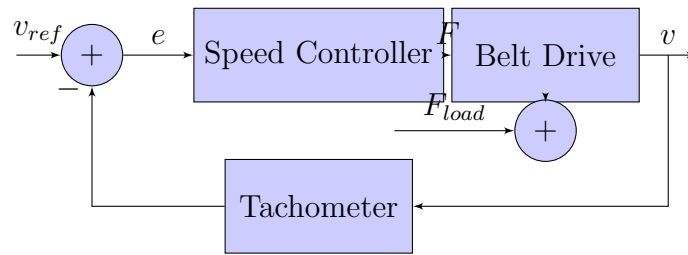


Figure 8: Conveyor Belt Control System Block Diagram

6 Linear Actuator System

6.1 System Description

A linear actuator with mass M , damping B , and spring constant k representing mechanical compliance and stiffness.

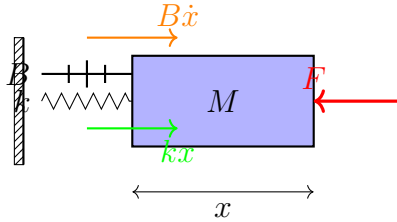
6.2 Governing Equation

$$M\ddot{x} + B\dot{x} + kx = F \quad (12)$$

where:

- M = mass (kg)
- B = damping coefficient (N·s/m)
- k = spring constant (N/m)
- F = applied force (N)
- x = position (m)

6.3 Free Body Diagram



Linear Actuator (Mass-Spring-Damper)

Figure 9: Free Body Diagram of Linear Actuator System

6.4 Transfer Function

$$G(s) = \frac{X(s)}{F(s)} = \frac{1}{Ms^2 + Bs + k} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} \quad (13)$$

where $\omega_n = \sqrt{k/M}$ and $\zeta = \frac{B}{2\sqrt{kM}}$

6.5 Control System Block Diagram

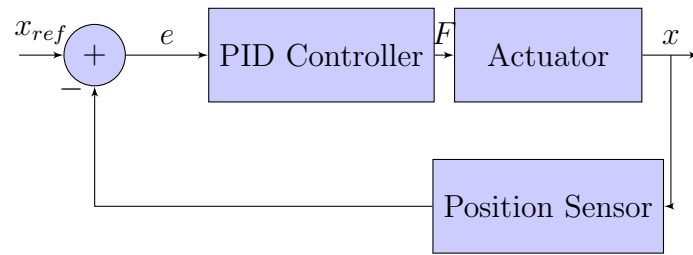


Figure 10: Linear Actuator Control System Block Diagram

7 Wheel Drive System

7.1 System Description

A wheel drive system with mass M , rolling resistance B , and traction force F for vehicle motion.

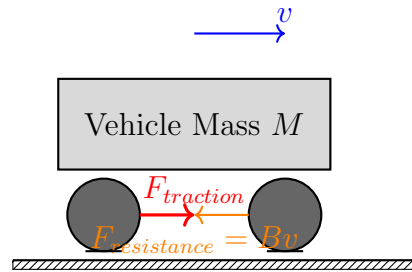
7.2 Governing Equation

$$M\dot{v} + Bv = F \quad (14)$$

where:

- M = vehicle mass (kg)
- B = rolling resistance coefficient (N·s/m)
- F = traction force (N)
- v = velocity (m/s)

7.3 Free Body Diagram



Wheel Drive Vehicle

Figure 11: Free Body Diagram of Wheel Drive System

7.4 Transfer Function

$$G(s) = \frac{V(s)}{F(s)} = \frac{1}{Ms + B} = \frac{K}{\tau s + 1} \quad (15)$$

where $K = 1/B$ and $\tau = M/B$

7.5 Control System Block Diagram

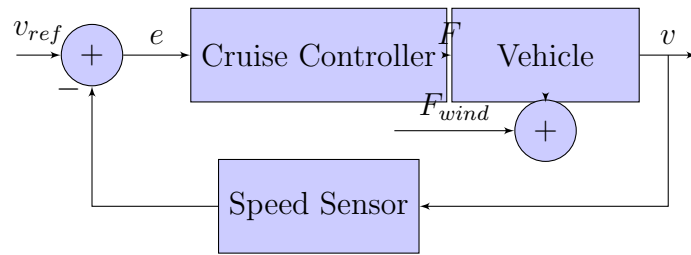


Figure 12: Wheel Drive Control System Block Diagram

8 Gearbox System

8.1 System Description

A gearbox with input and output inertias, gear ratio N , and various compliance and damping effects.

8.2 Governing Equations

Input side:

$$J_1 \ddot{\theta}_1 + B_1 \dot{\theta}_1 + K(\theta_1 - N\theta_2) = \tau_1 \quad (16)$$

Output side:

$$J_2 \ddot{\theta}_2 + B_2 \dot{\theta}_2 + K(\theta_2 - \frac{\theta_1}{N}) = \frac{\tau_L}{N} \quad (17)$$

where:

- J_1, J_2 = input and output inertias ($\text{kg}\cdot\text{m}^2$)
- B_1, B_2 = input and output damping ($\text{N}\cdot\text{m}\cdot\text{s}/\text{rad}$)
- K = shaft stiffness ($\text{N}\cdot\text{m}/\text{rad}$)
- N = gear ratio
- τ_1 = input torque, τ_L = load torque

8.3 Free Body Diagram

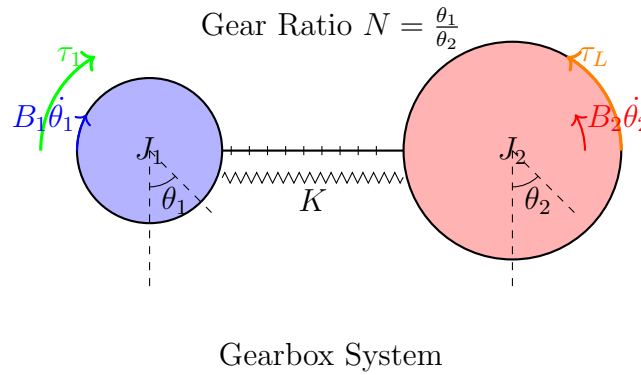


Figure 13: Free Body Diagram of Gearbox System

8.4 Control System Block Diagram

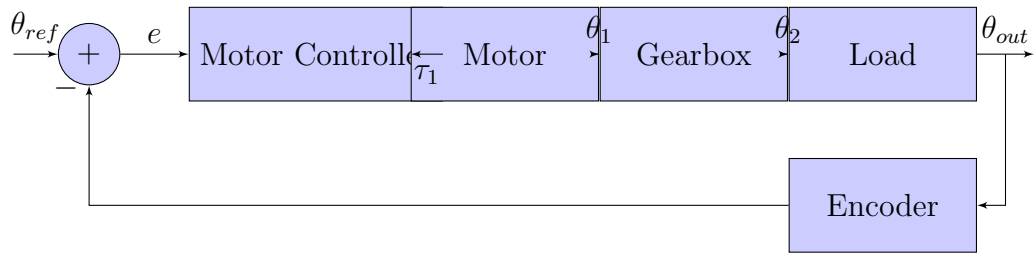


Figure 14: Gearbox Control System Block Diagram

9 Summary Table

Table 1: Summary of System Transfer Functions

System	Order	Transfer Function	Characteristics
Servo Motor	2nd	$\frac{K_t}{Js^2 + Ds}$	Integrator + 1st order
Robot Joint	2nd	$\frac{1}{Ms^2 + Bs + k}$	Second order
Ball Screw	2nd	$\frac{1}{s(Ms + B)}$	Integrator + 1st order
Conveyor Belt	1st	$\frac{1}{Ms + B}$	First order
Linear Actuator	2nd	$\frac{1}{Ms^2 + Bs + k}$	Second order
Wheel Drive	1st	$\frac{1}{Ms + B}$	First order
Gearbox	4th	Complex multi-body	Flexible coupling

10 Design Considerations

10.1 Stability Analysis

Each system's stability depends on pole locations:

- **First-order systems:** Always stable if $B > 0$
- **Second-order systems:** Stable if $B > 0$ and $k > 0$
- **Integrator systems:** Marginally stable, require feedback for stability

10.2 Performance Metrics

Key performance metrics for each system type:

- **Rise time:** $t_r \approx \frac{1.8}{\omega_n}$ for second-order systems
- **Settling time:** $t_s \approx \frac{4}{\zeta\omega_n}$ for second-order systems
- **Overshoot:** $MP = 100e^{-\frac{\pi\zeta}{\sqrt{1-\zeta^2}}}$ % for $\zeta < 1$
- **Steady-state error:** Depends on system type and input

10.3 Control Strategy Selection

- **PID Control:** Suitable for most systems with proper tuning
- **Lead/Lag Compensation:** For specific frequency domain requirements
- **State Feedback:** For multi-input multi-output systems
- **Adaptive Control:** For systems with parameter variations

11 Conclusion

This document provides comprehensive free body diagrams and control system representations for common mechanical systems. The mathematical models serve as the foundation for control system design, analysis, and implementation. Understanding these fundamental systems enables engineers to design effective control strategies for complex multi-body systems.